Table 9.2 Comparison of 5G, 6G, and satellite communication capabilities for drone operations.

Parameter	4G (legacy)	5G (current integration phase)	development)	networ
Latency	High (50–100 ms)	Very low (1–10 ms)	Ultra-low (<1 ms expected)	Modera (20–50 dependi orbit)
Bandwidth/data rate	Limited data capacity	High data capacity, suitable for HD video streams	Extremely high data capacity, supports multi-sensor + AI inference	Variable depends satellite bandwi and link stability
Network coverage	Urban and semi-urban areas	National networks; strong urban coverage	Global coverage envisioned with 6G mesh integration	Global, includir remote ocean regions
Drone swarm coordination ability	Low – delayed synchronization	High – supports real-time multi-UAV coordination	Very high – autonomous swarms with AI-distributed control	Modera depend line-of- and rela nodes
BVLOS flight support	Unreliable without relay systems	Supported with stable command-and-control links	Optimized for seamless BVLOS and autonomous routing	Strong i remote regions; weaker dense u areas
Integration with UAM/eVTOL systems	Not feasible	Feasible with supportive UTM systems	Fully integrated, core	Supplen support long-ra